MATH 3312, SPRING 2026: HOMEWORK 9

Definition 1. If R is a commutative ring, M is an R-module and $a \in R$, then we set

$$a \cdot M = aM = \{a \cdot m : \ m \in M\}.$$

This is an R-submodule of M (why?).

(1) Let R be a PID. Then, for $a, b \in R$, show that the map $R \xrightarrow{r \mapsto ar+(b)} R/(b)$ yields an isomorphism

$$R/(c) \xrightarrow{\simeq} a \cdot (R/(b)),$$

where $c = b/\gcd(a,b)$.[1]

Definition 2. Let R be a commutative ring and let M be an R-module. A **chain of submodules of length** n of an R-module M is a sequence

$$M_n < M_{n-1} < \cdots < M_1$$

of *non-zero* submodules of M where the $<$ denotes *strict* inclusion (that is, M_i is a *proper* submodule of M_{i-1}).

M has **length** n for $n \geq 1$ if there is a chain of submodules of length n, but no chain of length $n+1$. In this case, we will set $\mathrm{length}(M) = n$. If $M = \{0\}$, we will set $\mathrm{length}(M) = 0$. In all these cases, we will say that M has **finite length**.

(2) Suppose that M is an R-module of finite length. If $N \leq M$ is an R-submodule, show that N and M/N also have finite length, and that in fact, we have

$$\mathrm{length}(N) + \mathrm{length}(M/N) = \mathrm{length}(M).$$

(3) If R is a PID and $q \in R$ is an irreducible element, show that $R/(q^m)$ has length m as an R-module, and conclude that $q^r \cdot (R/q^m)$ has length $\max(0, m-r)$.

(4) Suppose that $p(x) \in \mathbb{Z}[x]$ is an irreducible element, and that $h_1(x), h_2(x) \in \mathbb{Z}[x]$ are such that $h_1(x)h_2(x) \in (p(x))$.
  (a) Show that $p(x)$ is primitive (see Problem 7 on Homework 8) and that it is also irreducible as an element of $\mathbb{Q}[x]$.
  (b) Show that, after switching the indices if necessary, there exists $g(x) \in \mathbb{Q}[x]$ such that $h_1(x) = p(x)g(x)$.
  (c) Show that $g(x) \in \mathbb{Z}[x]$ and therefore that $p(x)$ is a *prime* element of $\mathbb{Z}[x]$.
  (d) Conclude that $\mathbb{Z}[x]$ is a UFD.
  Hint: The argument for part (a) is like that used for Gauss's lemma in Homework 8.

Definition 3. Suppose that k is a field and that $A \in M_{n\times n}(k)$. For each $\lambda \in k$, the **algebraic multiplicity** of λ is the largest non-negative integer $m_A^{\mathrm{alg}}(\lambda)$ such that $(x-\lambda)^{m_A^{\mathrm{alg}}(\lambda)}$ divides $\mathrm{ch}_A(x)$.

The **geometric multiplcity** of λ is the quantity

$$m_A^{\mathrm{geom}}(\lambda) = \dim_k(\ker(A - \lambda I_n)).$$

[1] Note that the gcd is only well-defined up to invertible elements.

(5) Suppose that $A \in M_{n\times n}(k)$ and that $\lambda \in k$. Show that we always have
$$m_A^{\mathrm{geom}}(\lambda) \leq m_A^{\mathrm{alg}}(\lambda)$$
and that the following are equivalent:
(a) $m_A^{\mathrm{alg}}(\lambda)$ is non-zero.
(b) $m_A^{\mathrm{geom}}(\lambda)$ is non-zero.
(c) λ is an eigenvalue.
Hint: For the first part, it might be useful to look at the elementary divisor decomposition.

Definition 4. A matrix $B \in M_{n\times n}(k)$ is in **Jordan canonical form** if there exist $\lambda_1, \ldots, \lambda_r \in k$ and integers $m_1, \ldots, m_r \in \mathbb{Z}_{\geq 1}$ such that A is a block diagonal matrix of the form
$$B = \begin{bmatrix} J_{\lambda_1}(m_1) & 0 & \cdots & 0 \\ 0 & J_{\lambda_2}(m_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J_{\lambda_r}(m_r) \end{bmatrix}$$
Here, the diagonal entries are the Jordan block matrices from Homework 7.

(6) Show that the following are equivalent:
(a) A is k-conjugate to a matrix in Jordan canonical form.
(b) $\mathrm{ch}_A(x)$ splits into a product of linear factors in $k[x]$.
(c) $\min_A(x)$ splits into a product of linear factors in $k[x]$.

(7) Show that the following are equivalent:
(a) A is diagonalizable.
(b) $\min_A(x)$ splits into a product of *distinct* linear factors in $k[x]$.
Hint: What does the minimal polynomial have to do with the Jordan canonical form? It's best to think module theoretically for such things.

(8) Suppose that L/k is a field extension and that $f(x), g(x) \in k[x]$ and $h(x) \in L[x]$ are non-zero polynomials such that $f(x) = h(x)g(x)$. Show that $h(x) \in k[x]$.

(9) Prove *Eisenstein's criterion*: If R is a PID, $q \in R$ is irreducible, and $f(x) = x^d + a_{d-1}x^{d-1} + a_{d-2}x^{d-2} + \cdots + a_1 x + a_0 \in R[x]$ is such that $q \mid a_i$ for all $i = 0, \ldots, d-2$ but $q^2 \nmid a_0$, then $f(x)$ is irreducible. Use this to verify that the following polynomials in $\mathbb{Q}[x]$ are irreducible:
(a) $x^3 - 20x + 15$.
(b) $x^3 + 3x^2 + 3x - 1$.
(c) $x^2 - x - 1$.
(d) $x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$.
Hint: If $f(x)$ admits a factorization, what can you say about the factors after quotienting R by (q)? Also, for a couple of the examples, you might want to play around with substitutions for x.

Definition 5. A polynomial $f(x) \in k[x]$ is **separable** if $\gcd(f(x), f'(x)) = 1$ (see Proposition in Lecture 9).

(10) Let $f(x) \in k[x]$ is an irreducible and separable polynomial of degree d. Show that the following are equivalent for an extension $k \leq L$:
(a) $f(x)$ splits into a product of linear factors in $L[x]$.
(b) $|\mathrm{Emb}_k(k[x]/(f(x)), L)| = d$.
See Lecture 5 for the definition of Emb_k.